FINITE MORPHISMS ARE INTEGRAL, AND LOCALLY FINITE TYPE INTEGRAL MORPHISMS ARE FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement consists of two sentences without enumeration; both are inside the counted statement span and the proof handles them together.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Finiteness and integrality may be checked after base change to an affine over the base. Lemma lemma-integral-local (used also at 9696-9702 in this chapter), used at proof lines 9439-9441. Reduces both assertions to the affine case.
- The scheme case: finite equals integral plus locally of finite type. Morphisms, Lemma morphisms-lemma-finite-integral, used at proof lines 9441-9443. Is the input that closes both sentences.
- Representability of finite and integral morphisms of algebraic spaces. used at proof lines 9439-9440. Makes the base change reduction available.

Source reference. Original lemma 03ZQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5229–5229. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01WJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2484–2484. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A finite morphism of algebraic spaces is integral. An integral morphism of algebraic spaces which is locally of finite type is finite.*

Proof. In both cases the morphism is representable, and you can check the condition after a base change by an affine scheme mapping into Y, see Lemmas 03ZQ. Hence this lemma follows from the same lemma for the case of schemes, see Morphisms, Lemma 01WJ. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.